Reading the Bank Cards

FraudNet-Synth dashboard · Overview tab · the seven attributes explained simply

The four bank cards, as shown on the dashboard

Attribute	Bank A	Bank B	Bank C	Bank D
1. Mode badge	Augment · CTGAN	Augment · CTGAN	Schema · LLM	Schema · LLM
2. Transactions	69,546	59,599	39,695	29,767
3. Real frauds	149 (0.214%)	116 (0.195%)	40 (0.101%)	26 (0.087%)
4. Training frauds	105	82	28	18
5. Synthetic rows passed	868 of 1,000	814 of 1,000	234 of 300	264 of 300
6. Gate pass rate	86.8%	81.4%	78.0%	88.0%
7. Fidelity	0.804	0.800	0.746	0.756

Numbers are copied from the dashboard; the reasons below are taken from the project's decision log (docs/DECISIONS.md, decision IDs such as D-014 in brackets).

1. Mode badge: Augment · CTGAN or Schema · LLM

What it is: which method the bank uses to create extra fake fraud examples.

- **Augment Mode · CTGAN** (Banks A, B): CTGAN is a pair of neural networks, a *forger* and a *detective*. It studies the bank's own real fraud rows and learns to produce similar ones. It needs enough real examples to learn from, so only the data-rich banks use it.
- **Schema Mode · LLM** (Banks C, D): too few fraud rows to train CTGAN. Instead an AI language model (via Groq) gets a description of the columns (the *schema*) plus **summary statistics only**, such as each column's range and average, and writes rows. **No real row is ever sent** (D-015).

> **Analogy**
> *A student with many past papers (A, B) learns by studying them. A student with almost none (C, D) asks a tutor, who knows only the syllabus and average marks, to write practice questions.*

Why it matters: this badge *is* the research gap. Each bank's generation method is chosen by how much fraud data it has.

2. Transactions: 69,546 / 59,599 / 39,695 / 29,767

What it is: the total number of card payments (genuine + fraud) in the bank's private data, called its **shard**.

How it was made: in Step 2 the global training data was split into four unequal pieces. Bank A got the most, Bank D the least, like one big city bank and one small rural bank. The global validation and test sets were set aside *before* this split.

Why it matters: in FedAvg, a bank with more rows gets **more weight** in the average, so Bank A influences the shared model more than Bank D.

3. Real frauds: 149 (0.214%) / 116 (0.195%) / 40 (0.101%) / 26 (0.087%)

What it is: how many of the bank's transactions are actually fraud, and what percentage that is. For example, Bank A: 149 of 69,546 = 0.214%; Bank D: 26 of 29,767 = 0.087%.

Two kinds of inequality (non-IID):

- **Quantity skew:** banks have different numbers of transactions.

- **Label skew:** banks have different fraud rates. Bank A's rate is about **2.5 times** Bank D's.

Why it matters: this is why A and B are called *data-rich* and C and D *data-poor*, and why Bank D does so badly alone (fraud F1 0.456 in the six-arm results).

4. Training frauds: 105 / 82 / 28 / 18

What it is: the number of real fraud rows the bank actually **uses to train**. It is smaller than "Real frauds" because each bank splits its own data **70 / 15 / 15** (D-008):

Bank	Train (70%)	Validation (15%)	Local test (15%)	Total
A	105	22	22	149
B	82	17	17	116
C	28	6	6	40
D	18	4	4	26

- **Train:** the model learns from this.
- **Validation:** used to choose the alert threshold, never for learning.
- **Local test:** used only for the final exam.

Analogy

*You study from 70% of the question bank, use 15% for mock tests, and keep 15% sealed for the real exam. Studying the exam questions would make the score fake. Mixing these up is called **data leakage**.*

Why it matters: CTGAN learns only from these training frauds, and the LLM's statistics also come only from training data. Bank D learns from just **18** fraud examples. That is the core problem the project tries to solve.

5. Synthetic rows passed: 868 of 1,000 / 814 of 1,000 / 234 of 300 / 264 of 300

What it is: how many fake fraud rows were **generated** (the second number) and how many **passed the validation gate** (the first number).

Why 1,000 for CTGAN but 300 for the LLM?

- **CTGAN: 1,000** (D-014). It runs free on the laptop, so a big pool lets the gate reject freely.
- **LLM: 300** (D-017). Each row costs about 426 tokens on Groq; 1,000 rows would need roughly 425,000 tokens per bank, far beyond the free daily limit. 300 is still about 10× the real fraud count of C and D.

Important viva point

Not all passed rows are used in training. The rule is a **1:1 ratio** (D-025): each bank adds as many synthetic rows as it has real training frauds.

Bank A: 105 real + 105 synthetic (sampled from its 868) · Bank D: 18 real + 18 synthetic (sampled from its 264).

Why 1:1? Using all 868 would make fake rows outnumber real ones about 8 to 1, and the model could learn the generator's habits instead of real fraud. 1:1 keeps real data at half the weight, with the same rule for every bank, so the comparison stays fair.

6. Gate pass rate: 86.8% / 81.4% / 78.0% / 88.0%

What it is: passed ÷ generated × 100. For example, A: 868 ÷ 1,000 = 86.8%; C: 234 ÷ 300 = 78.0%.

What the gate checks, for every row from both modes:

- **Schema / range (Pandera):** correct columns, no missing values, values within the real range (+10% margin), Class = 1.
- **Edge-clamping:** rejects rows with 3 or more values stuck at the exact real minimum or maximum, a CTGAN habit.
- **Privacy, DCR (distance to closest record):** rejects rows that are almost a copy of a real customer row.
- **Diversity:** rejects near-duplicates of a fake row already accepted (mode collapse).
- **Embedding check (sentence-transformers):** rejects rows whose text form is identical.

Analogy

*A quality inspector at a factory gate. Products from two factories (CTGAN and the LLM) go through the **same** inspection.*

How to read it: a higher pass rate does **not** mean better data. Bank D (LLM) has the highest pass rate, 88.0%, but its fidelity (0.756) is lower than the CTGAN banks' (about 0.80). Pass rate asks "did it break any rule?"; fidelity asks "does it look like real fraud?". A rate well below 100% shows the gate is really filtering, not rubber-stamping.

7. Fidelity: 0.804 / 0.800 / 0.746 / 0.756

What it is: a score from **0 to 1** (SDMetrics library) for how statistically similar the passed fake rows are to the bank's real fraud rows. 1 means a perfect match. It averages two checks:

- **Column Shapes:** does each column, on its own, have the right distribution? (For example, are fake amounts spread like real fraud amounts?)
- **Column Pair Trends:** do pairs of columns move together as in real data? (When V14 is low, is V17 also low, as in real fraud?)

Analogy

Forging a painting: Column Shapes asks whether you used the right colours; Column Pair Trends asks whether the colours sit in the right places relative to each other.

How to read the numbers:

- **CTGAN (A, B) about 0.80 > LLM (C, D) about 0.75.** Expected: CTGAN learned from real rows, while the LLM saw only averages and ranges, so it gets each column roughly right but misses the relationships between columns.
- The gate also requires a **minimum batch fidelity of 0.70** (D-024). All four banks passed.
- Fidelity is scored on the **whole batch**, not row by row, so it can accept or refuse a bank's batch, not single rows.

The whole card in one sentence (to say to the guide)

"Bank D has 29,767 transactions but only 26 frauds, and just 18 of those are for training. With so little data it uses the LLM, which saw only statistics, never real rows. It generated 300 fake fraud rows; 264 passed our quality gate (88%), with fidelity 0.756, lower than CTGAN's 0.80 because the LLM can't learn how columns relate. Of those 264, we add only 18 to training, one fake for every real fraud."